



Global dynamics of a class of 3-dimensional piecewise linear observable system with two zones

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ABSTRACT

Piecewise smooth differential systems have garnered increasing attention due to their broad applications across various fields. Notably, existing literature primarily focuses on planar piecewise smooth systems. In this paper, we investigate a class of 3-dimensional piecewise linear homogeneous observable systems divided into two zones by a plane. We first establish the existence and stability of invariant cones, then analyze the dynamics on these cones utilizing properties of the Poincaré half map. Finally, through Poincaré compactification, we derive the global phase portraits within the Poincaré ball for this 3-dimensional system.

1. Introductions

In nonlinear dynamical systems, piecewise linear systems represent a particularly intriguing class for several reasons. Firstly, these systems frequently arise in realistic nonlinear engineering models. In various engineering domains, piecewise linear systems prove to be highly practical; for instance, certain devices can be accurately modeled using piecewise linear differential systems, see [1,2]. In control theory, we often employ piecewise linear models to represent systems, which are widely used in applications such as simple electronic oscillators and nonlinear control systems, see [3,4]. Furthermore, some models are explicitly constructed using piecewise linear representations, while others can be approximated through piecewise linear models by taking limits, see [5–7]. Although piecewise linear systems may appear simplistic, they exhibit a rich array of dynamics that encapsulate the essential behaviors of general smooth nonlinear systems. This complexity underscores their significant theoretical value in the study of differential systems, see [8,9].

Let $\Sigma = \{(x_1, x_2, x_3) \mid x_1 = 0\}$ be the discontinuous plane and $\Sigma^\pm = \{(x_1, x_2, x_3) \mid \pm x_1 > 0\}$. Consider the following 3-dimensional piecewise linear differential system

$$\dot{\mathbf{x}} = \begin{cases} A^+ \mathbf{x} + \mathbf{b}^+, & x_1 \in \Sigma^+, \\ A^- \mathbf{x} + \mathbf{b}^-, & x_1 \in \Sigma^-, \end{cases} \quad (1)$$

where $A^\pm$ are 3×3 matrices, $\mathbf{b}^\pm \in \mathbb{R}^3$ and $\mathbf{x} = (x_1, x_2, x_3)^T \in \mathbb{R}^3$, the $\dot{\mathbf{x}}$ denotes the derivative of $\mathbf{x}$ with respect to time s . System (1) has been extensively studied, addressing various aspects such as the existence of periodic orbits, see [10–13], homoclinic orbits, see [14], invariant cones, see [15], and Hopf-like bifurcations, see [16–19].

Observable systems play a crucial role in control theory as they enable the inference of all necessary states from a system's outputs. This capability is essential for accurate state estimation, which directly impacts effective feedback control and system stability. Without observability, controllers may fail to respond appropriately to changes, resulting in degraded performance or instability. This concept is fundamental in various applications, including control systems, signal processing, robotics, aerospace, and power systems. For example, in power systems, the state of the system (typically bus voltages and phase angles) is crucial for monitoring and control. However, it is not feasible to measure the state directly at every point in the grid. Instead, measurements such as power flows, voltage magnitudes, and current injections are taken at various locations. Observability ensures that these measurements are sufficient to estimate the entire state of the system. 3-dimensional piecewise linear observable systems are applied in many real-world models, such as “Non-van der Pol Modified Wien Bridge”, “Colpitts Oscillator” and “Chua System” (for specific details see [3]).

Given that system (1) comprises 24 parameters, we adopt the following hypotheses to focus on a significant class of system (1):

(H1) The system (1) is continuous homogeneous system. That is, $\mathbf{b}^+ = \mathbf{b}^- = \mathbf{0}$, A^+ and A^- share the last two columns, i.e. $A^+ = A^- + (A^+ - A^-)\mathbf{e}_1\mathbf{e}_1^T$.

(H2) The continuous homogeneous system (1) is observable. That is, the observability matrix

$$\mathcal{O} = \begin{pmatrix} \mathbf{e}_1^T \\ \mathbf{e}_1^T A^- \\ \mathbf{e}_1^T (A^-)^2 \end{pmatrix}$$

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has full rank. For more explanations see [Appendix A](#).

(H3) The eigenvalues of matrices $A^\pm$ are $\lambda^\pm$ and $\alpha^\pm \pm i\beta^\pm$ with $\beta^\pm > 0$.

According to [20], suppose that system (1) satisfies (Hi), $i = 1, 2, 3$, then it can be deduced into the following normal forms:

$$\dot{\mathbf{x}} = \begin{cases} A^+ \mathbf{x}, & x_1 \geq 0, \\ A^- \mathbf{x}, & x_1 \leq 0, \end{cases} \quad (2)$$

with

$$A^\pm = \begin{pmatrix} 2\alpha^\pm + \lambda^\pm & -1 & 0 \\ 2\alpha^\pm \lambda^\pm + (\alpha^\pm)^2 + (\beta^\pm)^2 & 0 & -1 \\ \lambda^\pm((\alpha^\pm)^2 + (\beta^\pm)^2) & 0 & 0 \end{pmatrix}, \quad (3)$$

we denote $\iota^\pm = 2\alpha^\pm + \lambda^\pm$ are the traces of $A^\pm$.

System (2) has been thoroughly analyzed in [8,20,21]. The authors of [20] investigated the existence, uniqueness, and stability of invariant cones for system (2) using Poincaré half maps. They also conjectured the existence of a saddle-node bifurcation of two-zonal invariant cones for certain parameter choices. Subsequently, in [21], this conjecture was proven, and the local expressions of the hyper-surface where the saddle-node bifurcation occurs were derived. Continuing the work from [20], the authors provided a rigorous proof for a counterintuitive example presented in [8], demonstrating that despite predictions of asymptotic stability from the linear components, the singular point located at the discontinuous plane can be unstable for system (2). Additionally, they established sufficient conditions for the global asymptotic stability of system (2).

According to [22], two piecewise smooth vector fields Z and $\tilde{Z}$, defined in open sets V and $\tilde{V}$ with discontinuity planes $\Sigma \subset V$ and $\tilde{\Sigma} \subset \tilde{V}$ respectively, are considered Σ -topologically equivalent if there exists an orientation-preserving homeomorphism $h: V \rightarrow \tilde{V}$ that maps Σ to $\tilde{\Sigma}$ and preserves the orbits of Z under $\tilde{Z}$.

Global phase portraits are invaluable tools for studying the long-term dynamical behavior of differential systems. They provide crucial information regarding the presence of attractors, repellers, invariant manifolds or limit cycles for given parameter values. Therefore, the analysis of global phase portraits is one of the most important issues in the qualitative theory of differential systems. Several studies have contributed to the understanding of global phase portraits for planar piecewise linear systems, see [23–26] and references therein. However, results concerning the global phase portraits of higher-dimensional piecewise smooth systems remain quite rare. The objective of our paper is to investigate the global phase portraits of 3-dimensional piecewise linear differential system, which can be summarized as follows.

Theorem 1. Consider the 3-dimensional piecewise linear system (2) with $\alpha^\pm < 0$ and $\lambda^\pm < 0$. Let

$$\gamma^\pm = \frac{\alpha^\pm - \lambda^\pm}{\beta^\pm}. \quad (4)$$

If $\gamma^+ \cdot \gamma^- \geq 0$, then the phase portraits of a piecewise linear system (2) is Σ -topologically equivalent to one of the 11 phase portraits described in Fig. 1.

The organization of the remainder of the paper is as follows. In Section 2, we present the preliminary results, specifically addressing the Poincaré half map in Section 2.1. The existence and stability of invariant cones in Section 2.2. Section 3 discusses the dynamics on the invariant cones, which are divided into three parts: $\gamma^- > 0$, $\gamma^+ = \gamma^- = 0$ and $\gamma^+ \cdot \gamma^- = 0$ with $\gamma^+ + \gamma^- \neq 0$. In Section 4, we introduce Poincaré compactification for 3-dimensional piecewise polynomial systems and examine the dynamics of infinite singular points. Section 5 provides the proof of Theorem 1. Finally, Appendix A introduces observable linear systems, and Appendix B provides a detailed proof of Proposition 3.

2. Preliminary results

2.1. Poincaré half map

The fundamental results regarding the Poincaré half map have been analyzed in [8]. To describe these maps, the auxiliary function

$$\varphi_\gamma(\tau) = 1 - e^{\gamma\tau}(\cos \tau - \gamma \sin \tau) \quad (5)$$

introduced by Andronov in [27] is crucial, see Fig. 2. This function satisfies the property $\varphi_{-\gamma}(\tau) = \varphi_\gamma(-\tau)$, and for $\gamma > 0$, there exists a value $\hat{\tau} \in (\pi, 2\pi)$ such that $\varphi_\gamma(\hat{\tau}) = 0$.

Using the flow of the system $\dot{\mathbf{x}} = A^- \mathbf{x}$ with $x_1 \leq 0$, some points $\mathbf{p} \in \Sigma$ can be transformed into points $\mathbf{q} \in \Sigma$. This allows us to define a left Poincaré half map P^- such that $\mathbf{q} = P^-(\mathbf{p})$. Similarly, we can define the right Poincaré half map P^+ which transforms points $\mathbf{q} \in \Sigma$ into points $\mathbf{r} \in \Sigma$. Consequently, the overall Poincaré map P can be expressed as $P(\mathbf{p}) = P^+ \circ P^-(\mathbf{p}) = P^+(\mathbf{q}) = \mathbf{r}$, see Fig. 3.

Let $\mathbf{x}(s)$ denote the solution of the system $\dot{\mathbf{x}} = A^- \mathbf{x}$ with the initial point $\mathbf{p} = (0, y_0, z_0)$ at $s = 0$. We define the orbit passing through $\mathbf{p}$ as $\Gamma_{\mathbf{p}}$. Since $\mathbf{e}_1^T \dot{\mathbf{p}} = -y_0$, where $\mathbf{e}_1 = (1, 0, 0)^T$, it follows that if $y_0 > 0$, the orbit $\Gamma_{\mathbf{p}}$ crosses the discontinuous plane from Σ^+ to Σ^- . By computing the expression for $\mathbf{x}(s)$ and imposing that $\mathbf{q} = \mathbf{x}(\tau/\beta^-) = (0, y_1, z_1)$ is the first intersection point of the orbit $\Gamma_{\mathbf{p}}$ and the discontinuous plane Σ , we can derive the Poincaré half map P^- in parametric form as follows:

$$\begin{aligned} u_0(\tau) &:= \frac{z_0}{y_0} = \lambda^- + \beta^-[(\gamma^-)^2 + 1] \frac{e^{\gamma^- \tau} \sin(\tau)}{\varphi_{\gamma^-}(\tau)}, \\ u_1(\tau) &:= \frac{z_1}{y_1} = \lambda^- - \beta^-[(\gamma^-)^2 + 1] \frac{e^{-\gamma^- \tau} \sin(\tau)}{\varphi_{-\gamma^-}(\tau)}. \end{aligned} \quad (6)$$

Furthermore,

$$\frac{y_1}{y_0} = -\frac{\varphi_{-\gamma^-}(\tau)}{\varphi_{\gamma^-}(\tau)} e^{(\gamma^- + \frac{\alpha^-}{\beta^-})\tau}, \quad (7)$$

where $y_0 > 0$, $y_1 < 0$ and $\tau \in (0, \hat{\tau}^-)$ where $\hat{\tau}^-$ is the first positive solution of $\varphi_{|\gamma^-|}(\tau) = 0$.

Similarly, since $\mathbf{e}_1^T \dot{\mathbf{q}} = -y_1$ and $y_1 < 0$, the orbit $\Gamma_{\mathbf{q}}$ passing through $\mathbf{q}$ crosses the discontinuous plane from Σ^- to Σ^+ . By computing the solution of the system $\dot{\mathbf{x}} = A^+ \mathbf{x}$ with the initial point $\mathbf{q} = (0, y_1, z_1)$ at $s = 0$, and imposing that $\mathbf{r} = \mathbf{x}(\tau/\beta^+) = (0, y_2, z_2)$ is the first intersection point of the orbit $\Gamma_{\mathbf{q}}$ and the discontinuous plane Σ , we can express the Poincaré half map P^+ in parametric form as follows:

$$\begin{aligned} u_1(\tau) &:= \frac{z_1}{y_1} = \lambda^+ + \beta^+[(\gamma^+)^2 + 1] \frac{e^{\gamma^+ \tau} \sin(\tau)}{\varphi_{\gamma^+}(\tau)}, \\ u_2(\tau) &:= \frac{z_2}{y_2} = \lambda^+ - \beta^+[(\gamma^+)^2 + 1] \frac{e^{-\gamma^+ \tau} \sin(\tau)}{\varphi_{-\gamma^+}(\tau)}. \end{aligned} \quad (8)$$

Furthermore,

$$\frac{y_2}{y_1} = -\frac{\varphi_{-\gamma^+}(\tau)}{\varphi_{\gamma^+}(\tau)} e^{(\gamma^+ + \frac{\alpha^+}{\beta^+})\tau}, \quad (9)$$

where $y_1 < 0$, $y_2 > 0$ and $\tau \in (0, \hat{\tau}^+)$ where $\hat{\tau}^+$ is the first positive solution of $\varphi_{|\gamma^+|}(\tau) = 0$.

From the above analysis, we have

Lemma 1. The following statements hold.

$$\frac{y_1(\tau)}{y_0(\tau)} = \begin{cases} \frac{u_0(\tau) - \lambda^-}{u_1(\tau) - \lambda^-} e^{\frac{\lambda^-}{\beta^-} \tau}, & \tau \in (0, \hat{\tau}^-), \tau \neq \pi, \\ -e^{\frac{\alpha^-}{\beta^-} \tau}, & \tau = \pi. \end{cases} \quad (10)$$

$$\frac{y_2(\tau)}{y_1(\tau)} = \begin{cases} \frac{u_1(\tau) - \lambda^+}{u_2(\tau) - \lambda^+} e^{\frac{\lambda^+}{\beta^+} \tau}, & \tau \in (0, \hat{\tau}^+), \tau \neq \pi, \\ -e^{\frac{\alpha^+}{\beta^+} \tau}, & \tau = \pi. \end{cases} \quad (11)$$

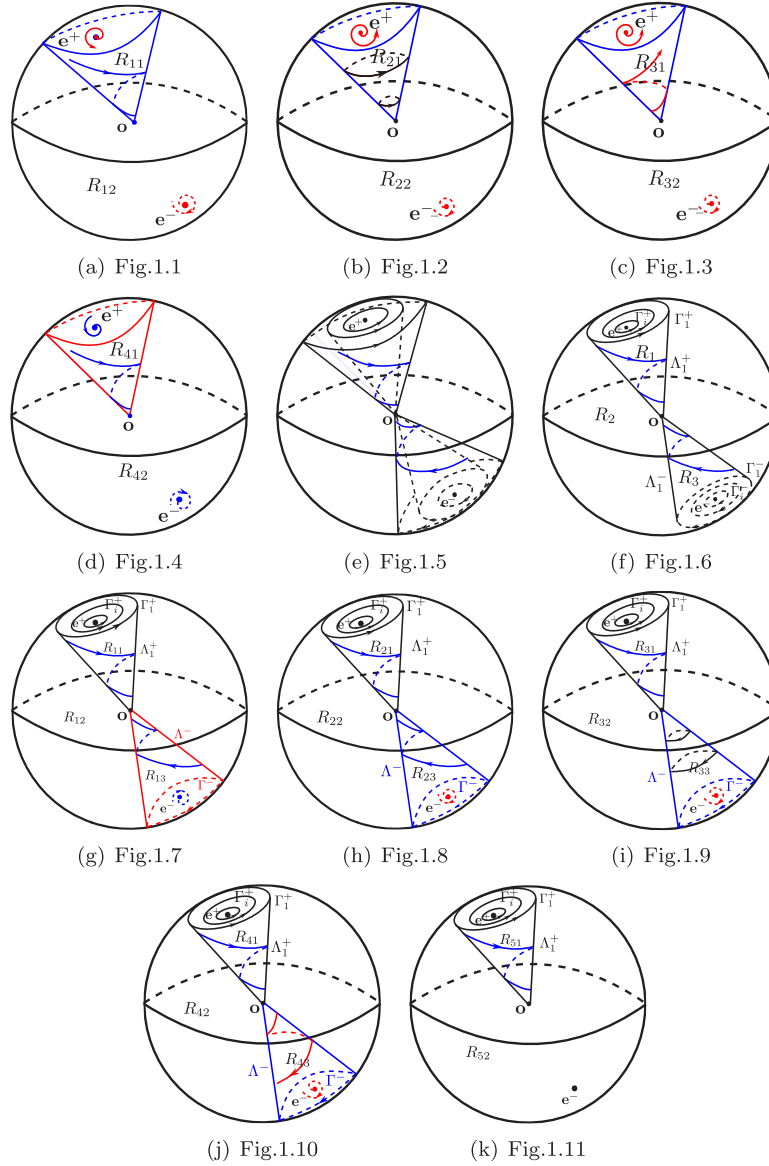


Fig. 1. Phase portraits of system (2).

2.2. The existence and stability of the invariant cones

In this section, we will use the Poincaré half maps to analyze the dynamics of system (2).

Definition 1. We say a manifold Λ is a cone if for all $x \in \Lambda$ we have that $\mu x \in \Lambda$ for every $\mu \geq 0$. Furthermore, the cone Λ is an invariant cone if there exists a semi-straight line contained in Σ that crosses the origin and is invariant under the Poincaré map P .

This cone will be referred to as a two-zonal invariant cone if it exists in two linear zones (see Fig. 4), and as a one-zonal invariant cone if it resides solely within one zone.

Furthermore, we can define a left slope transition map $u_1 = S^-(u_0)$, which transforms the slope u_0 of line l_0 into the slope u_1 of line l_1 . Similarly, a right slope transition map $u_2 = S^+(u_1)$ transforms the slope u_1 of line l_1 into the slope u_2 of line l_2 . The composite slope transition map can also be defined as $u_2 = S(u_0)$. It is evident that on the two-zonal invariant cone, we have $u_2 = S(u_0) = u_0$, see Fig. 4. Therefore, the existence of a two-zonal invariant cone corresponds to the existence of a fixed point of the composite slope transition map S .

Lemma 2. If there exists a fixed point $\bar{u}$ such that $S(\bar{u}) = S^+ \circ S^-(\bar{u}) = \bar{u}$, then we have a two-zonal invariant cone. Furthermore, the existence of fixed points of the composite slope transition map S corresponds with the existence of solutions to the equations:

$$\begin{aligned} u_0(\tau^-) &= u_2(\tau^+), \\ u_1(\tau^-) &= u_1(\tau^+). \end{aligned} \quad (12)$$

This is also equivalent to the existence of $S^-(\bar{u}) = (S^+)^{-1}(\bar{u})$.

Definition 2. We say a two-zonal invariant cone for system (2) is stable (resp. unstable) when the corresponding fixed point of $S^+ \circ S^-$ is stable (resp. unstable).

Next, we will introduce the existence and stability of the invariant cones for system (2). For the detail proof see [20].

Lemma 3. Assume that system (2) with $\gamma^+ \cdot \gamma^- \geq 0$, then the following statements hold.

- (a) If $\gamma^+ \cdot \gamma^- > 0$, then system (2) has a unique two-zonal invariant cone, which is hyperbolic and the following cases arise:
 - (a1) If $\gamma^+ > 0$ and $\gamma^- > 0$, the invariant cone is asymptotically stable.

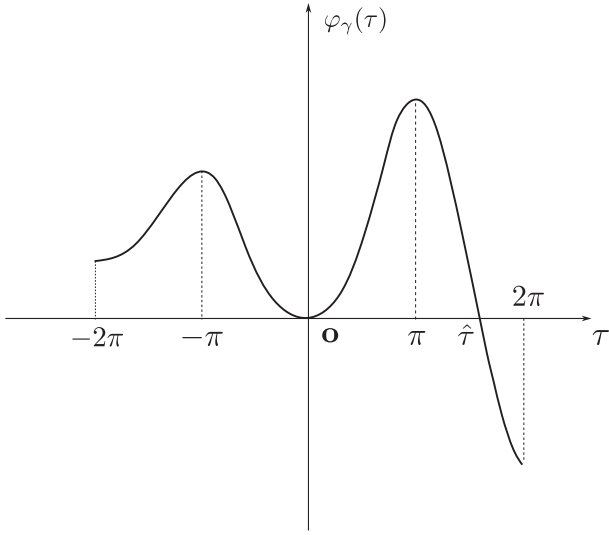


Fig. 2. The graph of the function $\varphi_\gamma(\tau)$ in (5) for $\gamma > 0$. Note that $\varphi_\gamma(\hat{\tau}) = 0$.

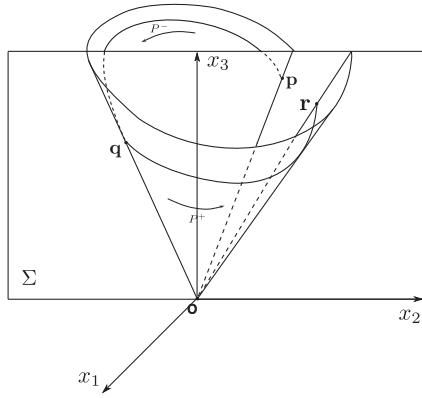


Fig. 3. Poincaré half maps P^+ and P^- of system (2).

- (a2) If $\gamma^+ < 0$ and $\gamma^- < 0$, the invariant cone is unstable.
- (b) If $\gamma^+ = \gamma^- = 0$, then the following statements hold:
 - (b1) If $\tau^+ = \tau^-$, then system (2) is foliated by invariant cones.
 - (b2) If $\tau^+ \neq \tau^-$, then system (2) has two continua of one-zonal invariant cones and no two-zonal invariant cones.
- (c) If $\gamma^+ \cdot \gamma^- = 0$ and $\gamma^+ + \gamma^- \neq 0$, then system (2) has a continuum of one-zonal invariant cones and the following cases arise:
 - (c1) If $\gamma^+ = 0$ and $(\tau^+ - \tau^-)\gamma^- < 0$, then system (2) has a unique two-zonal invariant cone, which is hyperbolic. Moreover the invariant cone is asymptotically stable when $\gamma^- > 0$ and unstable when $\gamma^- < 0$.
 - (c2) If $\gamma^+ = 0$ and $(\tau^+ - \tau^-)\gamma^- \geq 0$, then system (2) has no two-zonal invariant cones.
 - (c3) If $\gamma^- = 0$ and $(\tau^+ - \tau^-)\gamma^+ > 0$, then system (2) has a unique two-zonal invariant cone, which is hyperbolic. Moreover the invariant cone is asymptotically stable when $\gamma^+ > 0$ and unstable when $\gamma^+ < 0$.
 - (c4) If $\gamma^- = 0$ and $(\tau^+ - \tau^-)\gamma^+ \leq 0$, then system (2) has no two-zonal invariant cones.

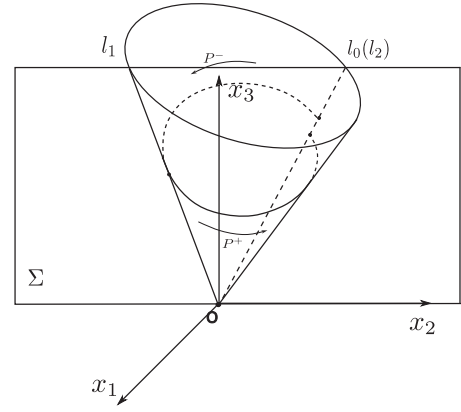


Fig. 4. The graph of a two-zonal invariant cone. Semi-straight l_0 is transformed into l_2 under the Poincaré map P , where l_0, l_1, l_2 are the intersections between the invariant zone and the discontinuous plane Σ .

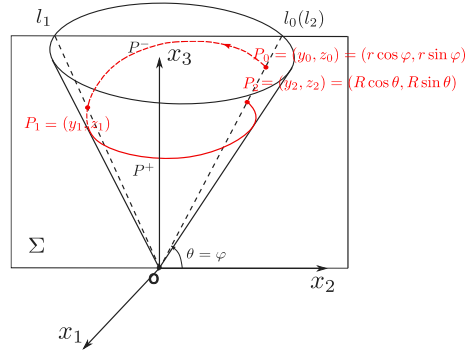


Fig. 5. The dynamics on the invariant cone when $\frac{R}{r} < 1$, where l_0 and l_1 are the intersection of the invariant cone and the discontinuous plane Σ .

3. The dynamics on the invariant cones

In this section, we examine the dynamics on the invariant cones. In [8], the authors investigated the global asymptotic stability of system (2) under certain conditions, outlined as follows:

Lemma 4. Assume that system (2) with $\alpha^\pm < 0$ and $\lambda^\pm < 0$ satisfies one of the following conditions:

- (a) System (2) has no two-zonal invariant cones.
- (b) The inequalities $\gamma^+ < 0$ and $\gamma^- < 0$ hold.
- (c) Both matrices share the real eigenvalue, that is, $\lambda^+ = \lambda^-$.
- (d) Both matrices share the complex eigenvalues, that is, $\alpha^+ = \alpha^-$ and $\beta^+ = \beta^-$.

Then the origin of system (2) is globally asymptotically stable.

For system (2), the authors in [20] considered the existence and stability of invariant cones. In Section 3 of this paper, we primarily focus on the dynamics on the invariant cones. Based on Lemma 4, we can ascertain that the origin of system (2) is globally asymptotically stable on the invariant cones for the subcases (a2, b2, c2, c4) outlined in Lemma 3. It is important to note that the dynamics on the invariant cones in the remaining subcases (a1, b1, c1, c3) described in Lemma 3 remain unknown, and this constitutes the first significant problem to be addressed in this paper.

By introducing polar coordinates and considering that $(y_1, z_1) = P^-(y_0, z_0)$ and $(y_2, z_2) = P^+(y_1, z_1)$, the dynamics on the invariant cones can be related to the ratio of the distances from the origin. This approach allows us to analyze how the system evolves on the invariant

cones based on these distances.

$$\frac{R}{r} = \sqrt{\frac{y_2^2 + z_2^2}{y_0^2 + z_0^2}} = \frac{y_2}{y_0} = \frac{y_1}{y_0} \frac{y_2}{y_1}, \quad (13)$$

Thus, we observe stable focus behavior when $\frac{R}{r} < 1$, a center when $\frac{R}{r} = 1$, and unstable focus behavior when $\frac{R}{r} > 1$, see Fig. 5.

According to the signs of $\gamma^\pm$, we can divide this section into three subsections.

3.1. The dynamics on the invariant cones with $\gamma^+ \cdot \gamma^- > 0$

Proposition 1. Assume that $\gamma^+ \cdot \gamma^- > 0$, then system (2) with $\alpha^\pm < 0$ and $\lambda^\pm < 0$ has a unique two-zonal invariant cone. The following statements hold:

- (i) If $\gamma^+ + \gamma^- > 0$, the dynamics on the invariant cone are either of stable focus type, or a center, or of unstable focus type.
- (ii) If $\gamma^+ + \gamma^- < 0$, the dynamics on the invariant cone are of stable focus type.

Proof. When $\gamma^+ + \gamma^- > 0$, this case has been proven in [8]. For the scenario where $\gamma^+ + \gamma^- < 0$, the conclusion can be easily obtained from statement (b) in Lemma 4.

3.2. The dynamics on the invariant cones with $\gamma^+ = \gamma^- = 0$

Proposition 2. Assume that $\gamma^+ = \gamma^- = 0$ and $\alpha^\pm < 0$, $\lambda^\pm < 0$. The following statements hold:

- (i) If $t^+ = t^-$, then $\mathbb{R}^3$ is foliated by invariant cones of system (2), and the dynamics on these invariant cones are of stable focus type.
- (ii) If $t^+ \neq t^-$, then system (2) has two continua of one-zonal invariant cones and no two-zonal invariant cones, with the dynamics on these invariant cones also being of stable focus type.

Proof. (i) Since $\gamma^+ = \gamma^- = 0$ and $t^+ = t^-$, we have $\alpha^+ = \lambda^+$, $\alpha^- = \lambda^-$ and $2\alpha^+ + \lambda^+ = 2\alpha^- + \lambda^-$. Consequently, we obtain $\lambda^+ = \lambda^-$. According to statement (c) in Lemma 4, we conclude that the dynamics on these invariant cones are of stable focus type.

(ii) The conclusion can be easily inferred from statement (a) in Lemma 4.

3.3. The dynamics on the invariant cones with $\gamma^+ \cdot \gamma^- = 0$ and $\gamma^+ + \gamma^- \neq 0$

Proposition 3.

Assume that $\gamma^+ \cdot \gamma^- = 0$ and $\gamma^+ + \gamma^- \neq 0$, then system (2) with $\alpha^\pm < 0$ and $\lambda^\pm < 0$ possesses a continuum of one-zonal invariant cones, with dynamics on these cones being of stable focus type. The following cases arise:

- (i) If $\gamma^+ = 0$ and $(t^+ - t^-)\gamma^- < 0$, then system (2) has a unique two-zonal invariant cone. The dynamics on this invariant cone are of stable focus type when $\gamma^- < 0$, and is either of stable focus type, a center, or of unstable focus type when $\gamma^- > 0$.
- (ii) If $\gamma^+ = 0$ and $(t^+ - t^-)\gamma^- \geq 0$, then system (2) has no two-zonal invariant cones.
- (iii) If $\gamma^- = 0$ and $(t^+ - t^-)\gamma^+ > 0$, then system (2) has a unique two-zonal invariant cone. The dynamics on this invariant cone are of stable focus type when $\gamma^+ < 0$ and is either of stable focus type, a center, or of unstable focus type when $\gamma^+ > 0$.
- (iv) If $\gamma^- = 0$ and $(t^+ - t^-)\gamma^+ \leq 0$, then system (2) has no two-zonal invariant cones.

Proof. The proof of this proposition is relatively lengthy, so we have included it in Appendix B.

4. The dynamics of infinite singular points

In this section, we study the dynamics of infinite singular points of system (2). The exploration of orbits escaping to infinity has been a crucial approach for understanding the global dynamics of the system. In 1881, Poincaré initiated the study of polynomial differential systems in the plane $\mathbb{R}^2$ using the central projection of orbits onto a sphere $\mathbb{S}^2$, tangent to the plane at the north pole. This method, known as Poincaré compactification, allows for the analysis of dynamics in the vicinity of infinity. Since then, Poincaré compactification has been extended to various cases, including n -dimensional polynomial differential systems [28,29], Hamiltonian differential systems [30], quasi-homogeneous differential systems [31,32], and rational differential systems [33,34]. Besides, the authors of [35] developed the Poincaré compactification to n -dimensional locally Lipschitz continuous vector fields. First, we will introduce Poincaré compactification for 3-dimensional piecewise polynomial systems, with more details see [36, 37].

Considering the following 3-dimensional piecewise polynomial system

$$\begin{cases} \dot{\mathbf{x}} = X(\mathbf{x}) = (X_1(\mathbf{x}), X_2(\mathbf{x}), X_3(\mathbf{x})), & x_1 \geq 0, \\ \dot{\mathbf{x}} = Y(\mathbf{x}) = (Y_1(\mathbf{x}), Y_2(\mathbf{x}), Y_3(\mathbf{x})), & x_1 \leq 0, \end{cases} \quad (14)$$

with $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$.

For simplicity, we consider its equivalent 3-dim piecewise polynomial vector fields

$$Z(\mathbf{x}) = (X, Y) = \begin{cases} X(\mathbf{x}), & x_1 \geq 0, \\ Y(\mathbf{x}), & x_1 \leq 0. \end{cases} \quad (15)$$

The discontinuous plane $\Sigma := \{\mathbf{x} = (x_1, x_2, x_3) \mid x_1 = 0\}$ divides $\mathbb{R}^3$ into two regions: $\Sigma^\pm = \{\mathbf{x} = (x_1, x_2, x_3) \mid \pm x_1 > 0\}$. In Σ^+ (resp. Σ^-), the degree m (resp. l) of the vector field X (resp. Y) is defined as $m = \max\{\text{degree}(X_i), i = 1, 2, 3\}$ (resp. $l = \max\{\text{degree}(Y_i), i = 1, 2, 3\}$).

Consider the hyperplane $\Pi = \{\mathbf{x} = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4 \mid x_4 = 1\}$ in $\mathbb{R}^4$, and let $\mathbb{S}^3 := \{\mathbf{y} = (y_1, y_2, y_3, y_4) \in \mathbb{R}^4 \mid y_1^2 + y_2^2 + y_3^2 + y_4^2 = 1\}$ represent the Poincaré sphere in $\mathbb{R}^4$. The origin $\mathbf{o} \in \mathbb{R}^4$ serves as the center of the Poincaré sphere $\mathbb{S}^3$, while Π is tangent to $\mathbb{S}^3$ at the north pole. Define the sets $E^\pm := \{\mathbf{y} \in \mathbb{S}^3 \mid \pm y_4 > 0\}$ and $\mathbb{S}^2 := \{\mathbf{y} \in \mathbb{S}^3 \mid \pm y_4 = 0\}$, then E^+ , E^- and $\mathbb{S}^2$ are referred to as the northern hemisphere, south hemisphere and the equator of $\mathbb{S}^3$ respectively, see Fig. 6. Using the central projections $\pi_\pm(\mathbf{x}) = \frac{\pm(x_1, x_2, x_3, 1)}{4(\mathbf{x})}$, where

$4(\mathbf{x}) = \sqrt{1 + x_1^2 + x_2^2 + x_3^2}$, the points $\mathbf{x} \in \mathbb{R}^3$ can be mapped to points $\mathbf{y} \in \mathbb{S}^3 \setminus \mathbb{S}^2$. Specifically, $\mathbb{R}^3$ is identified with the northern hemisphere via $\pi_+(\mathbf{x})$ and the southern hemisphere via $\pi_-(\mathbf{x})$, with $\mathbb{S}^2$ corresponding to the infinity of $\mathbb{R}^3$. Through these central projections, piecewise polynomial vector fields $Z(\mathbf{x})$ can be extended to a vector field on $\mathbb{S}^3 \setminus \mathbb{S}^2 = E^+ \cup E^-$,

$$\overline{Z}(\mathbf{y}) = y_4 \begin{pmatrix} 1 - y_1^2 & -y_2 y_1 & -y_3 y_1 \\ -y_1 y_2 & 1 - y_2^2 & -y_3 y_2 \\ -y_1 y_3 & -y_2 y_3 & 1 - y_3^2 \\ -y_1 y_4 & -y_2 y_4 & -y_3 y_4 \end{pmatrix} \begin{pmatrix} Z_1 \\ Z_2 \\ Z_3 \\ Z_4 \end{pmatrix}, \quad (16)$$

where $Z_i = Z_i\left(\frac{y_1}{|y_4|}, \frac{y_2}{|y_4|}, \frac{y_3}{|y_4|}\right)$.

Since the degree of vector field X is m and the degree of vector field Y is l , we can extend analytically the vector fields $\overline{Z}(\mathbf{y})$ to the whole sphere $\mathbb{S}^3$ by

$$P(Z)(\mathbf{y}) = \begin{cases} y_4^{m-1} \overline{X}(\mathbf{y}), & y_1 \geq 0, \\ y_4^{l-1} \overline{Y}(\mathbf{y}), & y_1 \leq 0, \end{cases} \quad (17)$$

where $X_i = X_i\left(\frac{y_1}{|y_4|}, \frac{y_2}{|y_4|}, \frac{y_3}{|y_4|}\right)$ and $Y_i = Y_i\left(\frac{y_1}{|y_4|}, \frac{y_2}{|y_4|}, \frac{y_3}{|y_4|}\right)$. The extended vector field $P(Z)$ is referred to as the Poincaré compactification of Z on $\mathbb{S}^3$. Since $\mathbb{S}^3$ is a differentiable manifold, we can compute the expression for $P(Z)$ at infinity by considering six local charts (U_i, F_i)

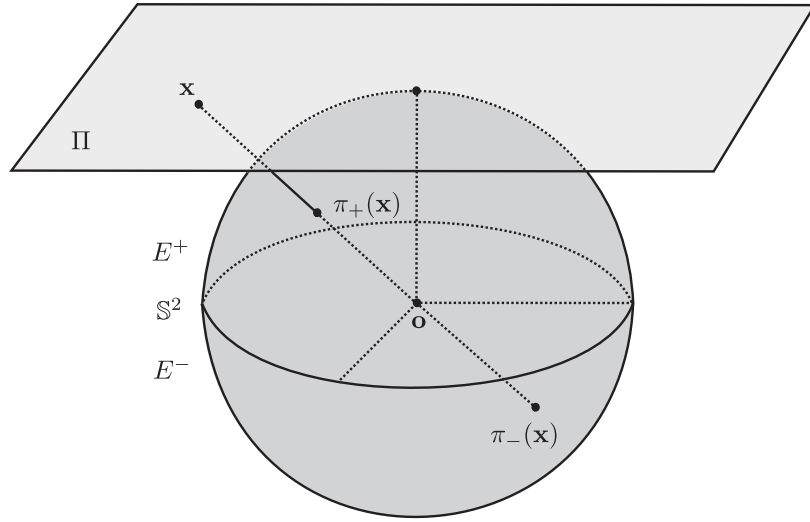
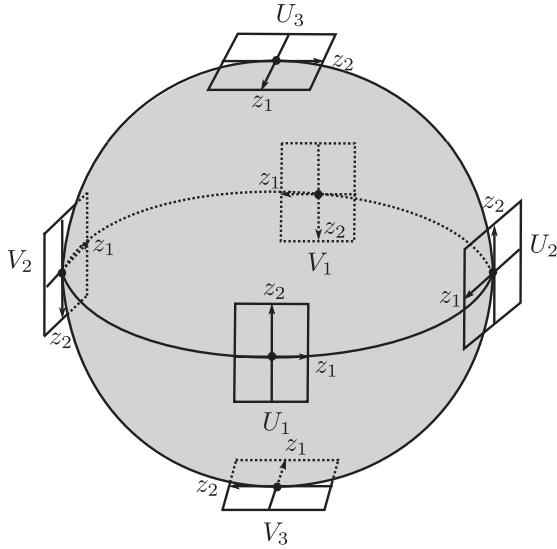


Fig. 6. The central projection of vector fields.

Fig. 7. The local charts U_i and V_i for $i = 1, 2, 3$ at infinity.

and (V_i, G_i) , where $U_i = \{y \in \mathbb{S}^3, y_i > 0\}$ and $V_i = \{y \in \mathbb{S}^3, y_i < 0\}$ for $i = 1, 2, 3$, see Fig. 7. The diffeomorphisms $F_i : U_i \rightarrow \mathbb{R}^3$ and $G_i : V_i \rightarrow \mathbb{R}^3$ for $i = 1, 2, 3$ serve as the inverses of the central projections from the origin to the tangent planes at the points $(\pm 1, 0, 0, 0)$, $(0, \pm 1, 0, 0)$ and $(0, 0, \pm 1, 0)$, respectively.

For the local chart U_1 . Doing the change of variables $x_1 = \frac{1}{z_3}, x_2 = \frac{z_1}{z_3}, x_3 = \frac{z_2}{z_3}$ with $z_3 > 0$, from (15) we have

$$\frac{z_3^m}{\Delta(\mathbf{z})^{m-1}}(X_2 - z_1 X_1, X_3 - z_2 X_1, -z_3 X_1), \quad (18)$$

where $X_i = X_i\left(\frac{1}{z_3}, \frac{z_1}{z_3}, \frac{z_2}{z_3}\right)$.

For the local chart V_1 . Doing the change of variables $x_1 = \frac{1}{z_3}, x_2 = \frac{z_1}{z_3}, x_3 = \frac{z_2}{z_3}$ with $z_3 < 0$, from (15) we have

$$\frac{(-1)^{l-1} z_3^l}{\Delta(\mathbf{z})^{l-1}}(Y_2 - z_1 Y_1, Y_3 - z_2 Y_1, -z_3 Y_1), \quad (19)$$

where $Y_i = Y_i\left(\frac{1}{z_3}, \frac{z_1}{z_3}, \frac{z_2}{z_3}\right)$.

Doing the change of variables $x_1 = \frac{z_1}{z_3}, x_2 = \frac{1}{z_3}, x_3 = \frac{z_2}{z_3}$ with $z_3 > 0$, from (15) we obtain the following piecewise polynomial vector fields in the local chart U_2 ,

$$\begin{cases} \frac{z_3^m}{\Delta(\mathbf{z})^{m-1}}(X_1 - z_1 X_2, X_3 - z_2 X_2, -z_3 X_2), & z_1 \geq 0, \\ \frac{z_3^l}{\Delta(\mathbf{z})^{l-1}}(Y_1 - z_1 Y_2, Y_3 - z_2 Y_2, -z_3 Y_2), & z_1 \leq 0, \end{cases} \quad (20)$$

and in the local chart V_2 ,

$$\begin{cases} \frac{(-1)^{m-1} z_3^m}{\Delta(\mathbf{z})^{m-1}}(X_1 - z_1 X_2, X_3 - z_2 X_2, -z_3 X_2), & z_1 \leq 0, \\ \frac{(-1)^{l-1} z_3^l}{\Delta(\mathbf{z})^{l-1}}(Y_1 - z_1 Y_2, Y_3 - z_2 Y_2, -z_3 Y_2), & z_1 \geq 0, \end{cases} \quad (21)$$

where $X_i = X_i\left(\frac{z_1}{z_3}, \frac{1}{z_3}, \frac{z_2}{z_3}\right)$ and $Y_i = Y_i\left(\frac{z_1}{z_3}, \frac{1}{z_3}, \frac{z_2}{z_3}\right)$.

Doing the change of variables $x_1 = \frac{z_1}{z_3}, x_2 = \frac{z_2}{z_3}, x_3 = \frac{1}{z_3}$ with $z_3 > 0$, from (15) we obtain the following piecewise polynomial vector fields in the local chart U_3 ,

$$\begin{cases} \frac{z_3^m}{\Delta(\mathbf{z})^{m-1}}(X_1 - z_1 X_3, X_2 - z_2 X_3, -z_3 X_3), & z_1 \geq 0, \\ \frac{z_3^l}{\Delta(\mathbf{z})^{l-1}}(Y_1 - z_1 Y_3, Y_2 - z_2 Y_3, -z_3 Y_3), & z_1 \leq 0, \end{cases} \quad (22)$$

and in the local chart V_3 ,

$$\begin{cases} \frac{(-1)^{m-1} z_3^m}{\Delta(\mathbf{z})^{m-1}}(X_1 - z_1 X_3, X_2 - z_2 X_3, -z_3 X_3), & z_1 \leq 0, \\ \frac{(-1)^{l-1} z_3^l}{\Delta(\mathbf{z})^{l-1}}(Y_1 - z_1 Y_3, Y_2 - z_2 Y_3, -z_3 Y_3), & z_1 \geq 0, \end{cases} \quad (23)$$

where $X_i = X_i\left(\frac{z_1}{z_3}, \frac{z_2}{z_3}, \frac{1}{z_3}\right)$ and $Y_i = Y_i\left(\frac{z_1}{z_3}, \frac{z_2}{z_3}, \frac{1}{z_3}\right)$.

From the above analysis, the infinity of 3-dimensional piecewise polynomial vector fields $Z(\mathbf{x})$, defined by (15), corresponds to $z_3 = 0$ of the vector fields in the local charts U_1, V_1, U_2, V_2, U_3 and V_3 , which are given in (18), (19), (20), (21), (22) and (23) respectively.

By using the above Poincaré compactification method, the dynamics of the infinity singular points for system (2) are summarized in the following proposition.

Proposition 4. The piecewise linear system (2) has two infinite singular points, $\mathbf{e}^\pm = (2\alpha^\pm, (\alpha^\pm)^2 + (\beta^\pm)^2, 0)$, located in the local charts U_1 and V_1 . Furthermore, $\mathbf{e}^\pm$ are stable foci when $\gamma^\pm < 0$, centers when $\gamma^\pm = 0$ and unstable foci when $\gamma^\pm > 0$.

Proof. In the local chart U_1 , doing the change of variables $z_1 = \frac{x_2}{x_1}$, $z_2 = \frac{x_3}{x_1}$ and $z_3 = \frac{1}{x_1}$ with $x_1 \geq 0$, then system (2) can be written as

$$\begin{cases} \dot{z}_1 = 2\alpha^+ \lambda^+ + (\alpha^+)^2 + (\beta^+)^2 - (2\alpha^+ + \lambda^+)z_1 + z_1^2 - z_2, \\ \dot{z}_2 = \lambda^+((\alpha^+)^2 + (\beta^+)^2) - (2\alpha^+ + \lambda^+)z_2 + z_1 z_2, \\ \dot{z}_3 = -z_3(2\alpha^+ + \lambda^+ - z_1). \end{cases} \quad (24)$$

Let $z_3 = 0$, $z_1 = 0$, we have $z_2 = z_1^2 - (2\alpha^+ + \lambda^+)z_1 + (\alpha^+)^2 + (\beta^+)^2 + 2\alpha^+ \lambda^+$. Substituting it into the expression for $\dot{z}_2$, we get $\dot{z}_2 = (z_1 - 2\alpha^+)((\beta^+)^2 + (-z_1 + \alpha^+ + \lambda^+))$. Let $\dot{z}_2 = 0$, we have either $z_1 = 2\alpha^+$ or $z_1 = (\alpha^+ + \lambda^+) \pm i\beta^+ \notin \mathbb{R}$. It is obvious that system (24) has a unique singular point $(z_1, z_2, z_3) = (2\alpha^+, (\alpha^+)^2 + (\beta^+)^2, 0)$. Consider the Jacobian matrix

$$J|_{(z_1, z_2)} = \begin{pmatrix} 2z_1 - 2\alpha^+ - \lambda^+ & -1 \\ z_2 & -2\alpha^+ - \lambda^+ + z_1 \end{pmatrix} \Big|_{(2\alpha^+, (\alpha^+)^2 + (\beta^+)^2)} \quad (25)$$

$$= \begin{pmatrix} 2\alpha^+ - \lambda^+ & -1 \\ (\alpha^+)^2 + (\beta^+)^2 & -\lambda^+ \end{pmatrix}.$$

The eigenvalues of matrix (25) are $\lambda_1 = (\alpha^+ - \lambda^+) + i\beta^+$ and $\lambda_2 = (\alpha^+ - \lambda^+) - i\beta^+$. Thus the singular point is a stable focus if $\alpha^+ - \lambda^+ < 0$, a center if $\alpha^+ - \lambda^+ = 0$ and an unstable focus if $\alpha^+ - \lambda^+ > 0$. Recall that $\gamma^+ = \frac{\alpha^+ - \lambda^+}{\beta^+}$, thus $\mathbf{e}^+$ is a stable focus if $\gamma^+ < 0$, a center if $\gamma^+ = 0$ and an unstable focus if $\gamma^+ > 0$.

In the local chart U_2 , let $z_1 = \frac{x_1}{x_2}$, $z_2 = \frac{x_3}{x_2}$ and $z_3 = \frac{1}{x_2}$, from (2) we have for $z_1 \geq 0$

$$\begin{cases} \dot{z}_1 = -1 + (2\alpha^+ + \lambda^+)z_1 + (2\alpha^+ \lambda^+ + (\alpha^+)^2 + (\beta^+)^2)z_1^2 - z_1 z_2, \\ \dot{z}_2 = \lambda^+((\alpha^+)^2 + (\beta^+)^2)z_1 - (2\alpha^+ \lambda^+ + (\alpha^+)^2 + (\beta^+)^2)z_1 z_2 + z_2^2, \\ \dot{z}_3 = -z_3((2\alpha^+ \lambda^+ + (\alpha^+)^2 + (\beta^+)^2)z_1 - z_2), \end{cases} \quad (26)$$

and for $z_1 \leq 0$

$$\begin{cases} \dot{z}_1 = -1 + (2\alpha^- + \lambda^-)z_1 + (2\alpha^- \lambda^- + (\alpha^-)^2 + (\beta^-)^2)z_1^2 - z_1 z_2, \\ \dot{z}_2 = \lambda^-((\alpha^-)^2 + (\beta^-)^2)z_1 - (2\alpha^- \lambda^- + (\alpha^-)^2 + (\beta^-)^2)z_1 z_2 + z_2^2, \\ \dot{z}_3 = -z_3((2\alpha^- \lambda^- + (\alpha^-)^2 + (\beta^-)^2)z_1 - z_2). \end{cases} \quad (27)$$

Since systems (26) and (27) have no singular point when $z_1 = z_3 = 0$, we can conclude that system (2) does not possess infinite singular points in the local chart U_2 . Additionally, it is important to note that the discontinuous plane $z_1 = 0$ represents a crossing region.

In the local chart U_3 , let $z_1 = \frac{x_1}{x_3}$, $z_2 = \frac{x_2}{x_3}$ and $z_3 = \frac{1}{x_3}$, from (2) we have for $z_1 \geq 0$

$$\begin{cases} \dot{z}_1 = (2\alpha^+ + \lambda^+)z_1 - z_2 - \lambda^+((\alpha^+)^2 + (\beta^+)^2)z_1^2, \\ \dot{z}_2 = -1 + (2\alpha^+ \lambda^+ + (\alpha^+)^2 + (\beta^+)^2)z_1 + \lambda^+((\alpha^+)^2 + (\beta^+)^2)z_1 z_2, \\ \dot{z}_3 = -\lambda^+((\alpha^+)^2 + (\beta^+)^2)z_1 z_3, \end{cases} \quad (28)$$

and for $z_1 \leq 0$

$$\begin{cases} \dot{z}_1 = (2\alpha^- + \lambda^-)z_1 - z_2 - \lambda^-((\alpha^-)^2 + (\beta^-)^2)z_1^2, \\ \dot{z}_2 = -1 + (2\alpha^- \lambda^- + (\alpha^-)^2 + (\beta^-)^2)z_1 + \lambda^-((\alpha^-)^2 + (\beta^-)^2)z_1 z_2, \\ \dot{z}_3 = -\lambda^-((\alpha^-)^2 + (\beta^-)^2)z_1 z_3. \end{cases} \quad (29)$$

Since the origin $(0, 0, 0)$ is not a singular point of systems (28) and (29), we can conclude that system (2) has no infinite singular points in the local chart U_3 . Direct computation reveals that the origin is a visible-invisible fold-fold point in this local chart.

Similarly, we can analyze the dynamics in the local charts V_i for $i = 1, 2, 3$. In the local chart V_1 , doing the change of variables $z_1 = \frac{x_2}{x_1}$,

Table 1

ω - and α -limit sets in the interior of each invariant region when $\gamma^+ > 0$ and $\gamma^- > 0$.

	R_{11}	R_{12}	R_{21}	R_{22}	R_{31}	R_{32}	R_{41}	R_{42}
ω -limit	$\mathbf{o}$	$\mathbf{o}$	Λ	Λ	Γ	Γ	$\mathbf{o}$	$\mathbf{o}$
α -limit	$\mathbf{e}^+$	$\mathbf{e}^-$	$\mathbf{e}^+$	$\mathbf{e}^-$	$\mathbf{o}$	$\mathbf{o}$	Γ	Γ

$z_2 = \frac{x_3}{x_1}$ and $z_3 = \frac{1}{x_1}$ with $x_1 \leq 0$, then system (2) can be written as

$$\begin{cases} \dot{z}_1 = 2\alpha^- \lambda^- + (\alpha^-)^2 + (\beta^-)^2 - (2\alpha^- + \lambda^-)z_1 + z_1^2 - z_2, \\ \dot{z}_2 = \lambda^-((\alpha^-)^2 + (\beta^-)^2) - (2\alpha^- + \lambda^-)z_2 + z_1 z_2, \\ \dot{z}_3 = -z_3(2\alpha^- + \lambda^- - z_1). \end{cases} \quad (30)$$

Letting $z_3 = 0$, similar to the analysis in U_1 , system (30) has a unique singular point $(z_1, z_2, z_3) = (2\alpha^-, (\alpha^-)^2 + (\beta^-)^2, 0)$ which is a stable focus if $\gamma^- < 0$, a center if $\gamma^- = 0$ and an unstable focus if $\gamma^- > 0$. And system (30) has no infinite singular points in the local charts V_2 and V_3 .

5. Proof of Theorem 1

In this section, we examine the global phase portraits of the piecewise linear system (2) and provide a proof for Theorem 1. We focus on the dynamics on the invariant cones by categorizing the signs of $\gamma^\pm$ into three distinct cases. The proof of Theorem 1 is derived from Propositions 5–7.

Proposition 5. For system (2) with $\gamma^+ \cdot \gamma^- > 0$, the following statements arise:

- If $\gamma^+ > 0$ and $\gamma^- > 0$, then the system has a unique invariant cone, characterized as a two-zone invariant cone. Furthermore, this invariant cone is stable. In this scenario, the dynamics on the invariant cone can exhibit behavior such as a stable focus, see Fig. 1.1, a center, see Fig. 1.2, or an unstable focus, see Fig. 1.3.
- If $\gamma^+ < 0$ and $\gamma^- < 0$, the system also has a unique two-zone invariant cone. However, in this case, the invariant cone is unstable, and the origin is globally asymptotically stable, see Fig. 1.4.

Proof. It is evident that system (2) possesses a unique finite singular point $\mathbf{o}$, two infinite singular points $\mathbf{e}^\pm$, and a unique two-zonal invariant cone Λ when $\gamma^+ \cdot \gamma^- > 0$. The invariant cone Λ intersects $\mathbb{S}^2$ in a periodic orbit Γ . We will structure the proof into the following two cases.

First, we consider case (i), where $\gamma^+ > 0$ and $\gamma^- > 0$. According to Lemma 3, the invariant cone Λ acts as an attractor. Additionally, Proposition 4 indicates that $\mathbf{e}^\pm$ are unstable foci. Based on statement (i) of Proposition 1, the dynamics on the invariant cone Λ can manifest as either a stable focus, a center or an unstable focus.

For subcase (i.1), the dynamics on the invariant cone Λ are of stable focus type. The invariant cone divides the Poincaré ball into two open regions, R_{11} and R_{12} , see Fig. 1.1. The ω - and α -limit sets are detailed in the second and third columns of Table 1.

For subcase (i.2), the dynamics on Λ correspond to a center. Here, the invariant cone separates the Poincaré ball into two open regions, R_{21} and R_{22} , see Fig. 1.2. The ω - and α -limit sets are presented in the fourth and fifth columns of Table 1.

For subcase (i.3), the dynamics on Λ are of an unstable type. The invariant cone again separates the Poincaré ball into two open regions, R_{31} and R_{32} , see Fig. 1.3. The ω - and α -limit sets are provided in the sixth and seventh columns of Table 1.

Secondly, we consider case (ii), where $\gamma^+ < 0$ and $\gamma^- < 0$. According to Lemma 3, the invariant cone Λ is repulsive. Additionally, Proposition 4 shows that $\mathbf{e}^\pm$ are stable foci. Based on Lemma 4(b), the origin is global asymptotically stable. From statement (ii) of Proposition 1, the dynamics on invariant cone Λ are of stable focus type. The invariant cone divides the Poincaré ball into two open regions, R_{41} and R_{42} , see Fig. 1.4. The ω - and α -limit sets are presented in the eighth and ninth columns of Table 1.

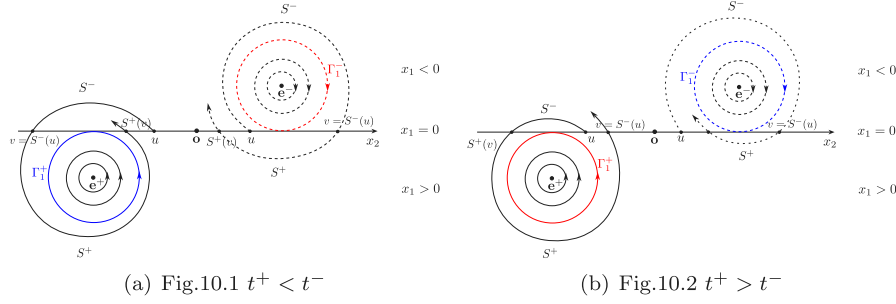


Fig. 8. The stability of periodic orbits with $\gamma^+ = \gamma^- = 0$ and $t^+ \neq t^-$.

Proposition 6. For system (2) with $\gamma^+ = \gamma^- = 0$, the following statements arise:

- (i) If $t^+ = t^-$, then $\mathbb{R}^3$ is foliated by invariant cones and the origin is globally asymptotically stable, see Fig. 1.5.
- (ii) If $t^+ \neq t^-$, the system exhibits two continua of one-zonal invariant cones and has no two-zonal invariant cones. The origin remains globally asymptotically stable. Furthermore, the one-zonal invariant cones $\Lambda_i^\pm$, which are tangent to the discontinuous plane, intersect $\mathbb{S}^2$ in periodic orbits $\Gamma_i^\pm$, see Fig. 1.6. The following cases arise:
 - (ii.1) If $t^+ < t^-$, the orbits outside Λ_1^+ are close to Λ_1^+ , while those outside Λ_1^- are far from Λ_1^- .
 - (ii.2) If $t^+ > t^-$, the orbits outside Λ_1^+ are far from Λ_1^+ , while those outside Λ_1^- are close to Λ_1^- .

Proof. We will structure the proof into the following two cases.

First, we consider case (i), where $t^+ = t^-$. Since $\gamma^+ = \gamma^- = 0$, we find that $\lambda^+ = \lambda^-$. According to statement (c) of Lemma 4, the origin of system (2) is globally asymptotically stable, see Fig. 1.5.

Secondly, we consider case (ii), where $t^+ \neq t^-$. According to statement (a) of Lemma 4, the origin of system (2) is globally asymptotically stable. Furthermore, the one-zonal invariant cones $\Lambda_i^\pm$, which are tangent to the discontinuous plane, intersect $\mathbb{S}^2$ in periodic orbits $\Gamma_i^\pm$. Additionally, the one-zonal invariant cones $\Lambda_i^\pm$ also intersect $\mathbb{S}^2$ in periodic orbits $\Gamma_i^\pm$ for $i \geq 2$, see Fig. 1.6. From [20], we can know if $\gamma^- = 0$, then $S^-(u) = -u + \frac{2}{3}t^-$. And if $\gamma^+ = 0$, then $S^+(v) = -v + \frac{2}{3}t^+$, where $v = S^-(u)$. Thus we can deduce

$$\begin{aligned}
 d(u) &= S^+ \circ S^-(u) - u \\
 &= S^+(v) - u \\
 &= -v + \frac{2}{3}t^+ - u \\
 &= -(-u + \frac{2}{3}t^-) + \frac{2}{3}t^+ - u \\
 &= \frac{2}{3}(t^+ - t^-).
 \end{aligned} \tag{31}$$

For subcase (ii.1), where $t^+ - t^- < 0$, we observe from Eq. (31) that $d(u) < 0$. In this scenario, the orbits outside Γ_1^+ are situated close to Γ_1^+ , while those outside Γ_1^- are located far from Γ_1^- on the sphere $\mathbb{S}^2$, see Fig. 8.1. Correspondingly, the orbits outside Λ_1^+ are close to Λ_1^+ , and the orbits outside Λ_1^- are distant from Λ_1^- . The invariant cones Λ_1^+ and Λ_1^- partition the Poincaré ball into three open regions: R_1 , R_2 and R_3 , see Fig. 1.6. The ω - and α -limit sets are detailed in the second, third and fourth columns of Table 2.

For subcase (ii.2), where $t^+ - t^- > 0$, we find from Eq. (31) that $d(u) > 0$. Consequently, the orbits outside Γ_1^+ are located far from Γ_1^+ , while those outside Γ_1^- are situated close to Γ_1^- on the sphere $\mathbb{S}^2$, see Fig. 8.2. Correspondingly, the orbits outside Λ_1^+ are far from Λ_1^+ , and the orbits outside Λ_1^- are close to Λ_1^- . The ω - and α -limit sets in regions R_1 and R_3 are consistent with those in subcase (ii.1). Additionally, the

Table 2

ω - and α -limit sets in the interior of each invariant region when $\gamma^+ = \gamma^- = 0$ and $t^+ < t^-$.

	R_1	R_2	R_3
ω -limit	$\mathbf{o}$	$\mathbf{o}$	$\mathbf{o}$
α -limit	Γ_1^+	Γ_1^-	Γ_1^-

ω -limit set in region R_2 is $\mathbf{o}$, while the α -limit set in R_2 is Γ_1^+ , see Fig. 1.6.

Proposition 7. For system (2) with $\gamma^+ \cdot \gamma^- = 0$ and $\gamma^+ + \gamma^- \neq 0$, there exists a continuum of one-zonal invariant cones, and the dynamics on these invariant cones are of stable focus type. Moreover, the following cases arise:

- (i) If $\gamma^+ = 0$, $\gamma^- < 0$ and $(t^+ - t^-) > 0$, then the system has a unique unstable two-zonal invariant cone. The dynamics on this invariant cone are of stable focus type, see Fig. 1.7.
- (ii) If $\gamma^+ = 0$, $\gamma^- > 0$ and $(t^+ - t^-) < 0$, the system has a unique stable two-zonal invariant cone. In this case, the dynamics on the invariant cone can manifest as a stable focus, see Fig. 1.8, a center, see Fig. 1.9, or an unstable focus, see Fig. 1.10.
- (iii) If $\gamma^+ = 0$ and $(t^+ - t^-)\gamma^- \geq 0$, the system does not possess a two-zonal invariant cone. The origin is globally asymptotically stable, see Fig. 1.11.
- (iv) If $\gamma^- = 0$, $\gamma^+ < 0$ and $(t^+ - t^-) < 0$, the system has a unique unstable two-zonal invariant cone. The dynamics on this invariant cone are of stable focus type, see Fig. 1.7.
- (v) If $\gamma^- = 0$, $\gamma^+ > 0$ and $(t^+ - t^-) > 0$, the system has a unique stable two-zonal invariant cone. In this case, the dynamics on the invariant cone can again be a stable focus, see Fig. 1.8, a center, see Fig. 1.9, or an unstable focus, see Fig. 1.10.
- (vi) If $\gamma^- = 0$ and $(t^+ - t^-)\gamma^+ \leq 0$, the system has no two-zonal invariant cone. The origin remains globally asymptotically stable, see Fig. 1.11.

Proof. We will structure the proof into the following six cases.

First, we consider case (i), where $\gamma^+ = 0$, $\gamma^- < 0$ and $(t^+ - t^-) > 0$. According to statement (c1) of Lemma 3, there exists a continuum of one-zonal invariant cones and an unstable two-zonal invariant cone Λ^- . Proposition 4 indicates that $\mathbf{e}^+$ is a center and $\mathbf{e}^-$ is a stable focus. Based on statement (i) of Proposition 3, the dynamics on the two-zonal invariant cone Λ^- are of stable focus type. Furthermore, the one-zonal invariant cones Λ_i^+ (resp. Λ^-) intersect $\mathbb{S}^2$ in periodic orbits Γ_i^+ (resp. Γ^-) for $i \geq 1$, where Γ_1^+ is tangent to the discontinuous plane Σ . The invariant cones Λ_1^+ and Λ^- partition the Poincaré ball into three open regions: R_{11} , R_{12} and R_{13} , see Fig. 1.7. The ω - and α -limit sets are provided in the second, third and fourth columns of Table 3.

Secondly, we consider case (ii), where $\gamma^+ = 0$, $\gamma^- > 0$ and $(t^+ - t^-) < 0$. According to statement (c1) of Lemma 3, there exists a continuum of one-zonal invariant cones and a stable two-zonal invariant cone Λ^- . Proposition 4 indicates that $\mathbf{e}^+$ is a center and $\mathbf{e}^-$ is an unstable focus.

Table 3

ω - and α -limit sets in the interior of each invariant region when $\gamma^+ = 0$ and $(t^+ - t^-)\gamma^- < 0$. ($i \geq 2$).

	R_{11}	R_{12}	R_{13}	R_{21}	R_{22}	R_{23}	R_{31}	R_{32}	R_{33}	R_{41}	R_{42}	R_{43}
ω -limit	$\mathbf{o}$	$\mathbf{o}$	$\mathbf{o}$	$\mathbf{o}$	$\mathbf{o}$	$\mathbf{o}$	$\mathbf{o}$	Λ^-	Λ^-	$\mathbf{o}$	Γ^-	Γ^-
α -limit	Γ_i^+	Γ^-	Γ^-	Γ_i^+	Γ_1^+	$\mathbf{e}^-$	Γ_i^+	Γ_1^+	$\mathbf{e}^-$	Γ_i^+	Γ_1^+	$\mathbf{o}$

Table 4

ω - and α -limit sets in the interior of each invariant region when $\gamma^+ = 0$ and $(t^+ - t^-)\gamma^- \geq 0$.

	$\gamma^- < 0$		$\gamma^- > 0$	
	R_{51}	R_{52}	R_{51}	R_{52}
ω -limit	$\mathbf{o}$	$\mathbf{o}$	$\mathbf{o}$	$\mathbf{o}$
α -limit	Γ_i^+	Γ_1^+	Γ_i^+	$\mathbf{e}^-$

Based on statement (i) of Proposition 3, the dynamics on the invariant cone Λ^- can be characterized as either a stable focus, a center or an unstable focus.

For subcase (ii.1), the dynamics on the invariant cone Λ^- are of stable focus type. The invariant cones Λ_1^+ and Λ^- partition the Poincaré ball into three open regions: R_{21} , R_{22} and R_{23} , see Fig. 1.8. The ω - and α -limit sets are presented in the fifth, sixth and seventh columns of Table 3.

For subcase (ii.2), the dynamics on the invariant cone Λ^- represent a center. The invariant cones Λ_1^+ and Λ^- partition the Poincaré ball into three open regions: R_{31} , R_{32} and R_{33} , see Fig. 1.9. The ω - and α -limit sets are provided in the eighth, ninth and tenth columns of Table 3.

For subcase (ii.3), the dynamics on the invariant cone Λ^- are of unstable focus type. The invariant cones Λ_1^+ and Λ^- partition the Poincaré ball into three open regions: R_{41} , R_{42} and R_{43} , see Fig. 1.10. The ω - and α -limit sets are presented in the eleventh, twelfth and thirteenth columns of Table 3.

Thirdly, we consider case (iii), where $\gamma^+ = 0$ and $(t^+ - t^-)\gamma^- \geq 0$. According to statement (c2) of Lemma 3, system (2) has no two-zonal invariant cones. From statement (a) of Lemma 4, the origin of system (2) is globally asymptotically stable. Since $\gamma^+ = 0$, it follows that $\mathbf{e}^+$ is a center. Furthermore, Proposition 4 indicates that if $\gamma^- < 0$, then $\mathbf{e}^-$ is a stable focus, while if $\gamma^- > 0$, $\mathbf{e}^-$ is an unstable focus. The invariant cone Λ_1^+ , which is tangent to the discontinuous plane, partitions the Poincaré ball into two open regions: R_{51} and R_{52} , see Fig. 1.11. The ω - and α -limit sets are provided in the second, third and fourth columns of Table 4.

Fourthly, we consider case (iv), where $\gamma^- = 0$, $\gamma^+ < 0$ and $(t^+ - t^-) < 0$. According to statement (c3) of Lemma 3, there exists a continuum of one-zonal invariant cones and an unstable two-zonal invariant cone Λ^+ . Proposition 4 indicates that $\mathbf{e}^-$ is a center and $\mathbf{e}^+$ is a stable focus. Based on statement (iii) of Proposition 3, the dynamics on the invariant cone Λ^+ are of stable focus type. The phase portraits of system (2) in this case can be derived by applying a symmetric transformation of case (i) with respect to the origin, see Fig. 1.7.

Fifthly, we consider case (v), where $\gamma^- = 0$, $\gamma^+ > 0$ and $(t^+ - t^-) > 0$. According to statement (c3) of Lemma 3, there exists a continuum of one-zonal invariant cones and a stable two-zonal invariant cone Λ^+ . Proposition 4 indicates that $\mathbf{e}^-$ is a center and $\mathbf{e}^+$ is an unstable focus. Based on statement (iii) of Proposition 3, the dynamics on the invariant cone Λ^+ can be characterized as either stable focus type, a center or unstable focus type. The phase portraits of system (2) in this case can be derived by applying a symmetric transformation of case (ii) with respect to the origin.

For the subcase (v.1), the dynamics on the invariant cone Λ^+ are of stable focus type, see Fig. 1.8.

For the subcase (v.2), the dynamics on the invariant cone Λ^+ represent a center, see Fig. 1.9.

For the subcase (v.3), the dynamics on the invariant cone Λ^+ are of unstable focus type, see Fig. 1.10.

Finally, we consider case (vi), where $\gamma^- = 0$ and $(t^+ - t^-)\gamma^+ \leq 0$. According to statement (c4) of Lemma 3, system (2) has no two-zonal invariant cones. From statement (a) of Lemma 4, the origin of system (2) is globally asymptotically stable. Since $\gamma^- = 0$, it follows that $\mathbf{e}^-$ is a center. Furthermore, Proposition 4 indicates that if $\gamma^+ < 0$, then $\mathbf{e}^+$ is a stable focus; if $\gamma^+ > 0$, $\mathbf{e}^+$ is an unstable focus. The phase portraits of system (2) in this case can be derived by applying a symmetric transformation of case (iii) with respect to the origin, see Fig. 1.11.

6. Conclusion

In this paper, we investigate a class of 3-dimensional piecewise linear systems with two zones separated by a plane. We present the main results regarding the global phase portraits of system (2) under certain hypotheses. First, we establish the existence and stability of invariant cones when $\gamma^+ \cdot \gamma^- \geq 0$. Utilizing the properties of the Poincaré half map and the slope map, we analyze the dynamics on these invariant cones. To achieve a comprehensive understanding of the global portraits, we also examine the dynamics of infinite singular points. Using the Poincaré compactification we obtained the global phase portraits of piecewise linear system (2). However, due to certain limitations in our conditions, we currently lack a method to determine the dynamics on the invariant cones when $\gamma^+ \cdot \gamma^- < 0$. This issue remains an area for future research.

CRedit authorship contribution statement

Qian Tong: Writing – original draft, Methodology, Formal analysis.
Shimin Li: Writing – review & editing, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Introduction to observable linear system

In control theory, a linear system is said to be “observable” if, by observing the output over a finite time interval, it is possible to determine the initial state of the system. Mathematically, consider a linear time-invariant (LTI) system described by:

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t), \end{aligned} \quad (\text{A.1})$$

where $\mathbf{x}(t)$ is the state vector, $\mathbf{y}(t)$ is the input vector and $\mathbf{u}(t)$ is the output vector. $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ are matrices of appropriate dimensions. The system is observable if the observability matrix $\mathcal{O}$ defined as:

$$\mathcal{O} = \begin{pmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \mathbf{C}\mathbf{A}^2 \\ \vdots \\ \mathbf{C}\mathbf{A}^{n-1} \end{pmatrix}$$

has full rank, where n is the dimension of the state vector $\mathbf{x}(t)$.

For (H2), since we assume that system (1) is a continuous homogeneous system, then it can be written as

$$\dot{\mathbf{x}} = \begin{cases} \mathbf{A}^+ \mathbf{x}, & x_1 \geq 0, \\ \mathbf{A}^- \mathbf{x}, & x_1 \leq 0, \end{cases} \quad (\text{A.2})$$

and matrices $A^\pm$ satisfy the continuity relation

$$A^+ = A^- + (A^+ - A^-)\mathbf{e}_1\mathbf{e}_1^T, \quad (\text{A.3})$$

where $\mathbf{e}_1$ is the first vector of the standard basis of $\mathbb{R}^3$. Then system (A.2) is equivalent to

$$\begin{aligned} \dot{\mathbf{x}} &= A^- \mathbf{x} + (A^+ - A^-)\mathbf{e}_1 u, \\ y &= \mathbf{e}_1^T \mathbf{x} = x_1, \end{aligned} \quad (\text{A.4})$$

where

$$u = \varphi(y) = \varphi(x_1) = \begin{cases} 0, & x_1 \leq 0, \\ x_1, & x_1 \geq 0. \end{cases} \quad (\text{A.5})$$

Actually, according to Eqs. (A.4) and (A.5), we have $\dot{\mathbf{x}} = A^- \mathbf{x}$ if $x_1 \leq 0$ and

$$\begin{aligned} \dot{\mathbf{x}} &= A^- \mathbf{x} + (A^+ - A^-)\mathbf{e}_1 x_1 \\ &= A^- \mathbf{x} + (A^+ - A^-)\mathbf{e}_1 \mathbf{e}_1^T \mathbf{x} \\ &= [A^- + (A^+ - A^-)\mathbf{e}_1 \mathbf{e}_1^T] \mathbf{x} \\ &= A^+ \mathbf{x} \end{aligned}$$

if $x_1 \geq 0$. Therefore, system (A.4) (or system (A.2)) is observable if the observability matrix

$$\mathcal{O} = \begin{pmatrix} \mathbf{e}_1^T \\ \mathbf{e}_1^T A^- \\ \mathbf{e}_1^T (A^-)^2 \end{pmatrix}$$

has full rank.

Appendix B. Proof of Proposition 3

The aim of this appendix is to show the proof of Proposition 3. Before that, we first need to introduce a continuous function $\Psi(\tau)$, and then utilize the properties of $\Psi(\tau)$ and maps $S^\pm$ to prove this proposition.

Lemma 5. Assume that $\alpha < 0$, $\lambda < 0$, and $\beta > 0$ and let $\hat{\tau}$ be the first positive zero of $\varphi_{|\gamma|}(\tau) = 0$, where $\gamma = (\alpha - \lambda)/\beta$. Then the continuous function

$$\Psi(\tau) = \begin{cases} \frac{\varphi_{-\gamma}(\tau)}{\varphi_{\gamma}(\tau)} e^{(\gamma + \frac{\alpha}{\beta})\tau}, & 0 < \tau < \hat{\tau}, \\ 1, & \tau = 0, \end{cases} \quad (\text{B.1})$$

satisfies the equality $\text{sign}(\Psi(\tau) - e^{\frac{\lambda}{\beta}\tau}) = \text{sign}(\gamma)$ for $0 < \tau < \hat{\tau}$.

Proof. The proof of Lemma 5 can be seen in [8].

To analyze the fixed points of the composite map $S^+ \circ S^-$, several properties of map S^- that follow from Proposition 15 in [20] are introduced in the next lemma. By observing the similarity in the expressions of parameters S^- and S^+ (see (6) and (8)), the properties of map S^+ is the same to S^- .

Lemma 6. For the slope transition map S^- the following statements are true.

- Map S^- satisfies $S^-(\lambda^-) = \lambda^-$ and the straight line $u_1 = -u_0 + \frac{2t^-}{3}$ is an oblique asymptote when $u_0 \rightarrow +\infty$.
- The map S^- is decreasing and its first derivative is given by $S^{-'}(u_0) = -e^{-4\gamma^- \tau^-} [\frac{\varphi_{\gamma^-}(\tau^-)}{\varphi_{-\gamma^-}(\tau^-)}]^3$ with $S^{-'}(\lambda^-) = -e^{\gamma^- \pi}$.
- If $\gamma^- = 0$, then $u_1 = S^-(u_0) = -u_0 + \frac{2t^-}{3}$.
- If $\gamma^- > 0$, then S^- is defined in $\mathbb{R}$ and $u_1 = u_H^- = \alpha^- - \beta^- \cot \hat{\tau}^-$ is an horizontal asymptote when $u_0 \rightarrow -\infty$.
- If $\gamma^- < 0$, then S^- is defined in $(u_v^-, +\infty)$, where $u_v^- = \alpha^- + \beta^- \cot \hat{\tau}^-$, and the straight line $u_0 = u_v^-$ is a vertical asymptote.
- If $\gamma^- \neq 0$, then the second derivative of map S^- satisfies $\gamma^- \cdot S^{-''}(u_0) < 0$.

Now, we will present the proof of Proposition 3 below.

Before examining the four cases in the proposition, we first prove the dynamics on the continuum of one-zonal invariant cones. We focus on the scenario where $\gamma^- = 0$ and $\gamma^+ \neq 0$, as the other situation is analogous. Since $\gamma^- = 0$, we only need to consider the left subsystem of (2).

$$\dot{\mathbf{x}} = A^- \mathbf{x}, \quad x_1 \leq 0 \quad (\text{B.2})$$

Note that $\gamma^- = 0$, we have $\alpha^- = \lambda^-$. In the following and for the sake of clarity, we will drop all the superscripts. The nonsingular matrix

$$P = \frac{1}{\beta} \begin{pmatrix} 0 & 1 & \beta \\ \beta & 2\alpha & 2\alpha\beta \\ \alpha\beta & \alpha^2 & \beta(\alpha^2 + \beta^2) \end{pmatrix}$$

reduce the matrix A to its Jordan canonical form

$$J = P^{-1}AP = \begin{pmatrix} \alpha & \beta & 0 \\ -\beta & \alpha & 0 \\ 0 & 0 & \alpha \end{pmatrix}.$$

It is obvious that

$$e^{Jt} = \begin{pmatrix} e^{\alpha t} \cos \beta t & e^{\alpha t} \sin \beta t & 0 \\ -e^{\alpha t} \sin \beta t & e^{\alpha t} \cos \beta t & 0 \\ 0 & 0 & e^{\alpha t} \end{pmatrix}.$$

Thus we have

$$e^{At} = P e^{Jt} P^{-1} = -\frac{e^{\alpha t}}{\beta^3} \begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix},$$

where

$$\begin{aligned} m_{11} &= (\alpha^2 \beta - \beta^3) \cos \beta t - 2\alpha \beta^2 \sin \beta t - \alpha^2 \beta, \\ m_{12} &= -\alpha \beta \cos \beta t + \beta^2 \sin \beta t + \alpha \beta, \\ m_{13} &= \beta \cos \beta t - \beta, \\ m_{21} &= 2\alpha^3 \beta \cos \beta t - (3\alpha^2 \beta^2 + \beta^4) \sin \beta t - 2\alpha^3 \beta, \\ m_{22} &= -(\beta^3 + 2\alpha^2 \beta) \cos \beta t + \alpha \beta^2 \sin \beta t + 2\alpha^2 \beta, \\ m_{23} &= 2\alpha \beta \cos \beta t + \beta^2 \sin \beta t - 2\alpha \beta, \\ m_{31} &= (\alpha^4 \beta + \alpha^2 \beta^3) \cos \beta t - (\alpha^3 \beta^2 + \alpha \beta^4) \sin \beta t - \alpha^2 \beta (\alpha^2 + \beta^2), \\ m_{32} &= -(\alpha \beta^3 + \alpha^3 \beta) \cos \beta t + \alpha \beta (\alpha^2 + \beta^2), \\ m_{33} &= \alpha^2 \beta \cos \beta t + \alpha \beta^2 \sin \beta t - \beta (\alpha^2 + \beta^2). \end{aligned}$$

Let $P_0 = (x_0, y_0, z_0)$ be any point on the invariant cones, the solution of system (B.2) at P_0 is

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = e^{At} \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} = \begin{pmatrix} -\frac{e^{\alpha t}}{\beta^3} (m_{11}x_0 + m_{12}y_0 + m_{13}z_0) \\ -\frac{e^{\alpha t}}{\beta^3} (m_{21}x_0 + m_{22}y_0 + m_{23}z_0) \\ -\frac{e^{\alpha t}}{\beta^3} (m_{31}x_0 + m_{32}y_0 + m_{33}z_0) \end{pmatrix}.$$

Since $\alpha < 0$, then $x_1(t) \rightarrow 0$, $x_2(t) \rightarrow 0$, $x_3(t) \rightarrow 0$ when $t \rightarrow +\infty$. Therefore, the dynamics on the continuum of one-zonal invariant cones are of stable focus type.

Next, we will prove the remaining four cases.

(i) If $\gamma^+ = 0$ and $(t^+ - t^-)\gamma^- < 0$, from statement (c1) of Lemma 3, then system (2) has a unique two-zonal invariant cone. From the analysis in the paragraph preceding Lemma 2, it follows that the existence of a two-zonal invariant cone corresponds to the existence of a fixed point for the composite slope transition map S . Therefore, there exists a unique fixed point for map S . According to Lemma 2, Eq. (12) has a unique solution, denoted as τ^+ and τ^- .

(i.1) When $\gamma^- < 0$, we have $t^+ > t^-$. According to Lemma 5, it follows that $\Psi(\tau^-) < e^{\frac{\lambda^-}{\beta^-} \tau^-}$ and $\Psi(\tau^+) = e^{\frac{\lambda^+}{\beta^+} \tau^+}$ under the conditions $\gamma^- < 0$, $\gamma^+ = 0$. Combining this with Eq. (13), we obtain

$$\frac{R}{r} = \frac{y_1}{y_0} \frac{y_2}{y_1} = \Psi(\tau^-) \Psi(\tau^+) < e^{\frac{\lambda^-}{\beta^-} \tau^-} e^{\frac{\lambda^+}{\beta^+} \tau^+} < 1.$$

Thus, the dynamics on the two-zonal invariant cone are of stable focus type.

(i.2) When $\gamma^- > 0$, we have $t^+ < t^-$. From Lemma 6, map S^- has the straight line $u_1 = -u_0 + \frac{2t^-}{3}$ as an oblique asymptote when $u_0 \rightarrow +\infty$,

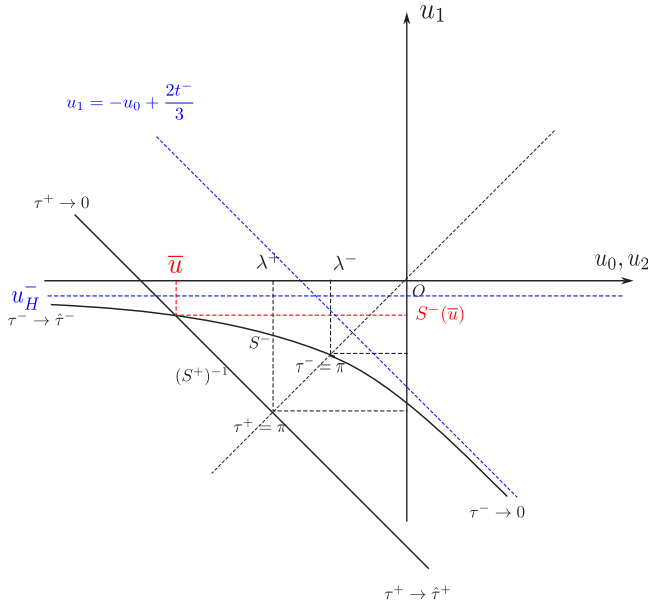


Fig. 9. Existence of invariant cones when $\gamma^+ = 0$, $\gamma^- > 0$ and $\lambda^+ < \lambda^-$. Note that $\tau^+ \in (0, \pi)$, $\tau^- \in (\pi, \hat{\tau}^-)$.

while S^+ becomes a straight line $u_2 = -u_1 + \frac{2t^+}{3}$. Given that $\gamma^- > 0$, $\gamma^+ = 0$ and $t^+ < t^-$, we obtain the inequalities $\frac{2t^-}{3} > 2\lambda^-$, $\frac{2t^+}{3} = 2\lambda^+$ and $\frac{2t^+}{3} < \frac{2t^-}{3}$.

For the case $\lambda^+ = \lambda^-$, we have $\tau^- = \tau^+ = \pi$ on the corresponding planar two-zonal invariant cone. By utilizing Eqs. (10), (11) and (13), we obtain

$$\frac{R}{r} = e^{\frac{\pi\alpha^+}{\beta^+} + \frac{\pi\alpha^-}{\beta^-}} < 1.$$

Thus, the dynamics on the planar two-zonal invariant cone are of stable focus type. By perturbing this situation, we can achieve similar dynamics on a non-planar two-zonal invariant cone. The concepts of the planar two-zonal invariant cone and the non-planar two-zonal invariant cone can be found in Remark 1.

For the case $\lambda^+ < \lambda^-$, we have $\frac{2t^+}{3} = 2\lambda^+ < 2\lambda^- < \frac{2t^-}{3}$, see Fig. 9. Under these conditions, we find that $\pi < \tau^- < \hat{\tau}^- < 2\pi$ for the phase τ^- of the Poincaré half map S^- at the fixed point $\bar{u}$, and $0 < \tau^+ < \pi$ for the phase τ^+ of the Poincaré half map S^+ at the fixed point $S^-(\bar{u})$. From Eq. (9) and Lemma 5, we have

$$\left| \frac{y_2}{y_1} \right| = \frac{\varphi_{-\gamma^+}(\tau^+)}{\varphi_{\gamma^+}(\tau^+)} e^{(\gamma^+ + \frac{\alpha^+}{\beta^+})\tau^+} = \Psi(\tau^+) = e^{\frac{\lambda^+}{\beta^+}\tau^+} > e^{\frac{\lambda^+}{\beta^+}\pi}. \quad (\text{B.3})$$

On the two-zonal invariant cone we have $u_0(\tau^-) = u_2(\tau^+) < \lambda^+ < \lambda^-$, $\lambda^- < u_1(\tau^-) = u_1(\tau^+) < u_H^-$. Combining with Eq. (10), we have

$$\left| \frac{y_1}{y_0} \right| = \left| \frac{u_0(\tau^-) - \lambda^-}{u_1(\tau^-) - \lambda^-} \right| e^{\frac{\lambda^-}{\beta^-}\tau^-} = \frac{\lambda^- - u_0(\tau^-)}{u_1(\tau^-) - \lambda^-} e^{\frac{\lambda^-}{\beta^-}\tau^-} > \frac{\lambda^- - \lambda^+}{u_H^- - \lambda^-} e^{\frac{\lambda^-}{\beta^-}2\pi}. \quad (\text{B.4})$$

Choosing $\beta^+ = -\lambda^+$ and imposing that $-\beta^- = \lambda^- > \lambda^+ + (u_H^- - \lambda^-)e^{3\pi}$, we can use Eqs. (13), (B.3) and (B.4) to conclude that

$$\frac{R}{r} > \frac{\lambda^- - \lambda^+}{u_H^- - \lambda^-} e^{\frac{\lambda^-}{\beta^-}2\pi} e^{\frac{\lambda^+}{\beta^+}\pi} > e^{3\pi} e^{-2\pi} e^{-\pi} = 1.$$

Thus, the orbits on the two-zonal invariant cone satisfy $R > r$, indicating that the dynamics on the two-zonal invariant cone are of unstable focus type under these conditions.

Finally, by combining the case of stable focus type when $\lambda^+ = \lambda^-$ and applying continuity, we find that there exists a specific value of λ^- such that $R = r$. In this scenario, the dynamics on the two-zonal invariant cone exhibit center behavior.

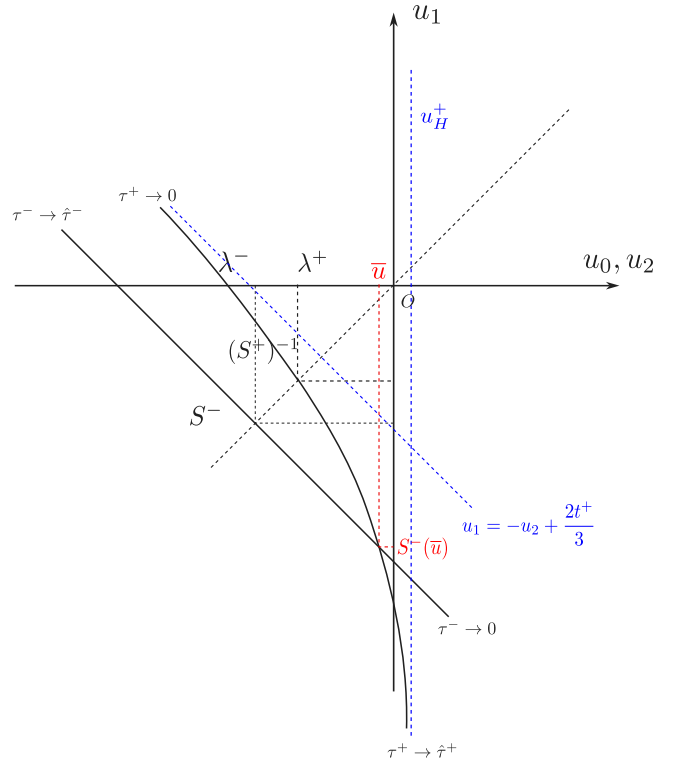


Fig. 10. Existence of invariant cones when $\gamma^- = 0$, $\gamma^+ > 0$ and $\lambda^- < \lambda^+$. Note that $\tau^- \in (0, \pi)$, $\tau^+ \in (\pi, \hat{\tau}^+)$.

(ii) System (2) has no two-zonal invariant cones but possesses a continuum of one-zonal invariant cones. According to statement (a) of Lemma 4, we can conclude that the dynamics on these one-zonal invariant cones are of stable focus type. The dynamics of invariant cones for case (iv) are similar to those in case (ii).

(iii) If $\gamma^- = 0$ and $(t^+ - t^-)\gamma^+ > 0$, the proof follows similarly to case (i). We can conclude that the dynamics on the two-zonal invariant cone are of stable focus type when $\gamma^+ < 0$. When $\gamma^+ > 0$, we can achieve the dynamics of a stable focus type with a non-planar two-zonal invariant cone by perturbing the scenario where $\lambda^+ = \lambda^-$.

To explore the possibility of unstable dynamics on the two-zonal invariant cone, we choose $\lambda^- < \lambda^+$, leading to the inequality $\frac{2t^-}{3} = 2\lambda^- < 2\lambda^+ < \frac{2t^+}{3}$, see Fig. 10. Similarly, from Eq. (7) and Lemma 5, we have

$$\left| \frac{y_1}{y_0} \right| = \frac{\varphi_{-\gamma^-}(\tau^-)}{\varphi_{\gamma^-}(\tau^-)} e^{(\gamma^- + \frac{\alpha^-}{\beta^-})\tau^-} = \Psi(\tau^-) = e^{\frac{\lambda^-}{\beta^-}\tau^-} > e^{\frac{\lambda^-}{\beta^-}\pi}. \quad (\text{B.5})$$

On the two-zonal invariant cone, we have $\lambda^- < \lambda^+ < u_0(\tau^-) = u_2(\tau^+) < u_H^+$, $u_1(\tau^-) = u_1(\tau^+) < \lambda^- < \lambda^+$. Combining with (11), we have

$$\left| \frac{y_2}{y_1} \right| = \left| \frac{u_1(\tau^+) - \lambda^+}{u_2(\tau^+) - \lambda^+} \right| e^{\frac{\lambda^+}{\beta^+}\tau^+} = \frac{\lambda^+ - u_1(\tau^+)}{u_2(\tau^+) - \lambda^+} e^{\frac{\lambda^+}{\beta^+}\tau^+} > \frac{\lambda^+ - \lambda^-}{u_H^+ - \lambda^+} e^{\frac{\lambda^+}{\beta^+}2\pi}. \quad (\text{B.6})$$

Choosing $\beta^- = -\lambda^-$ and imposing that $-\beta^+ = \lambda^+ > \lambda^- + (u_H^+ - \lambda^+)e^{3\pi}$, we can use Eqs. (13), (B.5) and (B.6) to conclude that

$$\frac{R}{r} > \frac{\lambda^+ - \lambda^-}{u_H^+ - \lambda^+} e^{\frac{\lambda^+}{\beta^+}2\pi} e^{\frac{\lambda^-}{\beta^-}\pi} > e^{-\pi} e^{3\pi} e^{-2\pi} = 1.$$

Therefore, under these conditions, the dynamics on the two-zonal invariant cone are of unstable focus type.

Finally, by combining the case of stable focus type when $\lambda^+ = \lambda^-$ and applying continuity, we find that there exists a specific value of λ^- such that $R = r$. In this scenario, the dynamics on the two-zonal invariant cone exhibit center behavior.

Remark 1. The focal plane of matrix A refers to the invariant plane of the system $\dot{\mathbf{x}} = A\mathbf{x}$ that includes the origin, where A is a real matrix with eigenvalues λ and $\alpha \pm i\beta$ ($\beta > 0$). This focal plane is also a planar invariant cone of the system. According to Lemma 19 from [20], the focal plane of

$$\text{matrix } M = \begin{pmatrix} 2\alpha + \lambda & -1 & 0 \\ 2\alpha\lambda + \alpha^2 + \beta^2 & 0 & -1 \\ \lambda(\alpha^2 + \beta^2) & 0 & 0 \end{pmatrix} \text{ is given by } \Pi_F \equiv \lambda^2 x - \lambda y + z =$$

0. For system (2), there are two focal half-planes, namely the left focal half-plane Π_{HF}^- and the right focal half-plane Π_{HF}^+ :

$$\Pi_{HF}^- \equiv (\lambda^-)^2 x - (\lambda^-)y + z = 0, \quad \mathbf{x} \leq 0,$$

and

$$\Pi_{HF}^+ \equiv (\lambda^+)^2 x - (\lambda^+)y + z = 0, \quad \mathbf{x} \geq 0.$$

Therefore, if $\lambda^+ = \lambda^-$, the two focal half-planes combine to form a planar two-zonal invariant cone. If system (2) has a non-planar two-zonal invariant cone (as shown in Fig. 4), this invariant cone is either above or below both whole focal plane Π_F^- and Π_F^+ . For more details, please refer to [20].

Data availability

No data was used for the research described in the article.

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